

Track 2 — Drug Repurposing for MVA1 (BUB1B compound heterozygosity)

Team: ciberpty · Rare Disease, Real Kid: MVA Hackathon 2026
Repository: <https://github.com/cpu-16/mva-hackathon-2026> · **Methods description:** `methods_track2.md`

Read this first — one page for the family and the treating team

Plain-language summary, placed first on purpose. The rest of this report is written for scientists; this page is written for the child's parents and for the clinician sitting with them. Every claim in it is sourced in the sections that follow.

What we looked for

An existing, already-approved medicine that could help a child with mosaic variegated aneuploidy caused by two changes in the *BUB1B* gene. We looked at every candidate the scientific literature points to, and we set out in advance the tests a candidate had to pass.

What we are not recommending

No drug should be started for this today. We want to be direct about that, because the honest answer is more useful than an encouraging one.

- **Metformin** is the medicine most often suggested for this kind of problem. It should not be started on this reasoning: no one has measured what dose would be needed to act on *this* problem, and the one route proposed for it — an effect on the cell's energy machinery — would need roughly a thousand times more drug than a person can safely take. (§3.1)
- **Bortezomib** is a real, approved cancer medicine that acts on the specific weakness these cells have. It reaches the needed

concentration. But it causes nerve damage in about 18 in 100 children who receive it, and this child already has muscle weakness. **Outside an active cancer, that trade is not worth making.** (§4)

What can be done this month

	Action	Why
1	Confirm the child's current age with the care team	Every schedule below depends on it, and we do not have it
2	Renal ultrasound every 3 months, birth to age 7	Published consensus recommendation for all forms of MVA (PMID 39264246) — not our idea, and not optional
3	Regular full clinical examination , including the orbit, skin and soft tissues	Rhabdomyosarcoma is not a kidney tumour and ultrasound will not find it. A 2026 case describes a child with MVA3 — the same condition caused by a different gene —
whose second tumour appeared behind the eye at age 12 (PMID 42595739)		
4	Avoid unnecessary radiation ; keep HPV vaccination up to date	Consensus guidance for these disorders (PMID 39264246)
5	Ask about testing and counselling for both parents	The clinical notes mention repeated miscarriages. In 2026 that pattern was linked to carriers of <i>BUB1B</i> changes, who showed measurable chromosome abnormalities in a blood test (PMID 42434306). This is a hypothesis, not a diagnosis — but it is a simple test, and it bears on any future pregnancy (§7)

What would change the answer

If a new tumour appears, the question changes from “should he take a medicine” to “which medicines belong in the treatment plan”. At that point bortezomib becomes worth putting to a tumour board, because

the comparison is no longer against nothing — it is against chemotherapy that carries its own harms. **It would not replace standard treatment.** It would be a question to ask, backed by the evidence in §3 and tested first in the laboratory experiment in §6.

We have written that answer down now so that nobody has to assemble it in a hurry later.

What we are not certain about

The strongest evidence for this medicine comes from cancer patients, not from children with MVA. Whether the same weakness exists in this child's cells is genuinely unknown, and §6 describes an experiment that could show us we are wrong. We would rather say that plainly than overstate what we have.

Executive summary

We applied five filters to candidate drugs rather than the usual two, and the third one changes the answer.

The candidate the literature points to is **metformin**. It descends from the AICAR hit in the aneuploidy-selective screen of Tang et al. (PMID 21315436), it is approved, and its mechanism is plausible. It fails on pharmacokinetics: plasma concentrations at therapeutic doses are **micromolar**, while complex I inhibition requires **millimolar** (PMID 37343530) — an gap of roughly **three orders of magnitude (~1000-fold)**. We expect metformin to be widely proposed in this track. **It cannot inhibit complex I at therapeutic plasma concentrations.** Whether it could still stress aneuploid cells through AMPK activation at those concentrations is a *different* claim, and an untested one: no study reports a metformin aneuploidy-selective effective concentration. We reject it as unevaluable on the mechanism it is proposed for, not as disproven — the distinction is set out in §3.1.

The candidate that survives the pharmacology is **bortezomib**. Aneuploid cells carry a stoichiometric protein imbalance and compensate by increasing protein degradation, which makes them **preferentially dependent on the proteasome**. Ippolito et al. (2024, *Cancer Discovery*, PMID 39247952) established this across multiple

diploid-versus-aneuploid models, recapitulated it in hundreds of cancer cell lines and primary tumours, and found that **aneuploidy level was significantly associated with the response of multiple myeloma patients to proteasome inhibitors** — a human clinical association rather than a pure cell-culture inference.

We state the size of the remaining leap rather than hide it. That evidence is drawn from **cancer** aneuploidy, and myeloma is a plasma-cell neoplasm carrying its own immunoglobulin-driven proteotoxic load. Whether the same dependency holds in **constitutional, variegated** aneuploidy is an extrapolation, and it is exactly what the experiment in §6 is designed to break (an eight-week assay window; about twelve weeks from a fresh biopsy).

We then tested that leap ourselves, and it did not survive. Before looking at any result we pre-registered a test on public data (§3.5): does the aneuploidy–proteasome link extend beyond haematological cancers? Across 444 solid-tumour cell lines no proteasome inhibitor reached significance, and across 946 lines the genetic dependency on **PSMB5 — bortezomib’s own target — is significantly weaker in solid tumours than in haematological ones** (interaction $p = 0.026$), with ploidy excluded as a confounder. This child’s tumour was a solid one. We report this because it weakens the argument we are making, and because we would rather find it than have a judge find it.

Bortezomib clears the pharmacokinetic filter with margin: EC₅₀ in highly aneuploid lines is below **40 nM**, while the approved 1.3 mg/m² IV dose reaches a C_{max} of **89–120 ng/mL = 231–312 nM** — a ratio of **5.8–7.8**. We quote the whole label interval rather than its ceiling; read it as an order-of-magnitude margin, not a measured number.

It then fails our fourth filter, and we say so plainly: **not for this child, not now**. Bortezomib causes peripheral neuropathy in 18% of paediatric patients (8% motor), and this child already has skeletal muscle atrophy. Outside an active malignancy that risk-benefit does not hold.

What remains is a specific and actionable proposition. Bortezomib does not repair the lesion — nothing restores BubR1 function — but it is the agent that exploits the best-characterised *consequence* of that lesion, and it is the one worth having ready for an MVA patient **with**

an active tumour, where the comparator is not “no drug” but conventional cytotoxic chemotherapy. This child is an embryonal rhabdomyosarcoma survivor with a high risk of embryonal tumours (PMID 28553959) — and he has already had one. That sequence is not hypothetical: a 2026 case report describes a girl with MVA3 who had a Wilms tumour at three and an orbital embryonal rhabdomyosarcoma at **twelve** (PMID 42595739). Our contribution is to have the answer ready — with its evidence, its concentration margin and its stopping rule — before the decision has to be made under pressure.

We also flag an antagonism a combination-minded team could walk into: **mTOR inhibitors would be expected to protect aneuploid cells from proteasome inhibition, not synergise** (PMID 31530568).

1. The variants and what they break

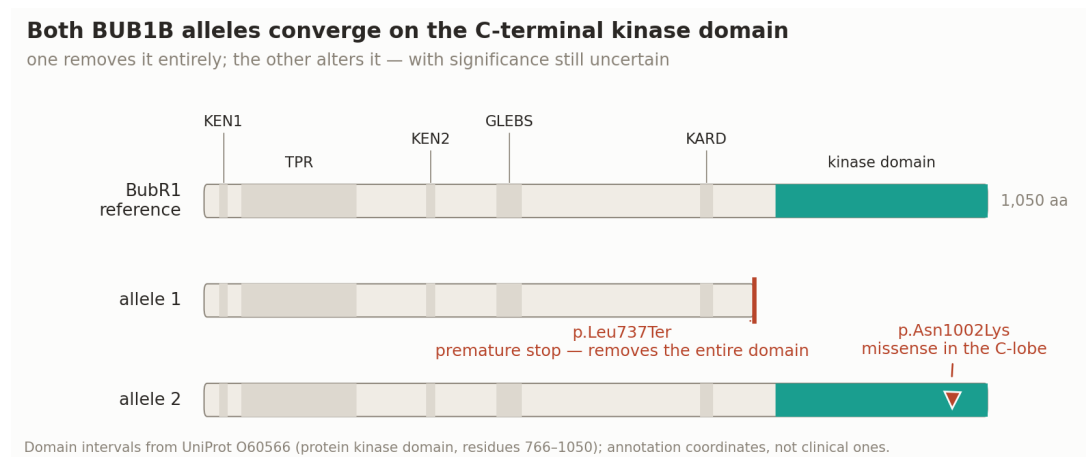
The proband is compound heterozygous in **BUB1B** (NM_001211.6, transcript ENST00000287598.11, MANE):

Variant	Consequence	Exomiser ACMG	ClinVar
c.2210T>G p. (Leu737Ter)	stop_gained	PATHOGENIC (PVS1, PM2_Supporting, PP4_Moderate, PP5_Strong)	533901 — Pathogenic/Likely pathogenic
c.3006T>G p. (Asn1002Lys)	missense	UNCERTAIN_SIGNIFICANCE (PM2_Supporting, PP4_Moderate, BP1)	not present

Population frequency depends on the source, and we report both rather than select the lower one. Exomiser’s gnomAD snapshot gives **9.98×10^{-5}** (exomes, NFE) for the nonsense and **9.0×10^{-7}** for the missense; Ensembl VEP returned 3×10^{-5} and no record respectively. Both are rare; neither is absent.

A note of caution on the missense. ClinVar holds **c.3006T>A**, a *different* nucleotide substitution producing the same p.Asn1002Lys, classified VUS. That classification does not transfer to our allele: PS1 applies when the known same-amino-acid variant is pathogenic, and here it is uncertain. Our variant is simply not in ClinVar.

BubR1 is a core component of the mitotic checkpoint complex and of kinetochore–microtubule attachment control; biallelic *BUB1B* mutation was the founding cause of constitutional aneuploidy with cancer predisposition (Hanks 2004, PMID 15475955), and BubR1 insufficiency in mice produces the progeroid phenotype that makes MVA an aging syndrome as well as a cancer syndrome (PMID 15208629).



Domain map of BubR1 showing both alleles converging on the C-terminal kinase domain

p.Leu737Ter truncates in the linker N-terminal to the C-terminal kinase fold — UniProt O60566 annotates the protein kinase domain at residues **766–1050** — removing it entirely, about 313 of the protein’s 1050 residues. (Whether BubR1 is catalytically active in humans is contested; we call it the kinase domain because that is the current UniProt annotation, and our argument rests on the truncation, not on catalysis.) It lands 16 residues upstream of **BUBR1^{ΔX753}**, a characterised truncating MVA allele (PMID 31738183) — very nearly the same truncation, not merely a related model.

p.Asn1002Lys falls inside that same domain, in its C-lobe, outside KEN, TPR and GLEBS, and introduces a positive charge. **No published functional assay of this substitution exists.**

The closest characterised missense sits ten residues away, and the mouse work on it carries a warning for us. Sieben et al. modelled the human MVA missense **BUBR1^{ΔL1012P}** in mice and paired it with **BUBR1^{ΔX753}** — the same truncating-plus-missense architecture as our proband, with the missense in the same domain. Those **BubR1^{ΔX753/}**

L1012P animals **died prematurely**; the viable genotype paired the missense with a *hypomorphic* allele that still yields some wild-type protein (PMID 31738183).

Our proband is alive carrying a truncating allele. If that result transfers, it constrains p.Asn1002Lys: **it would have to retain more residual function than L1012P does** — a hypomorph rather than a functional null. That is a falsifiable prediction about our own VUS, and the allele-fate arm of the experiment in §6 tests it directly.

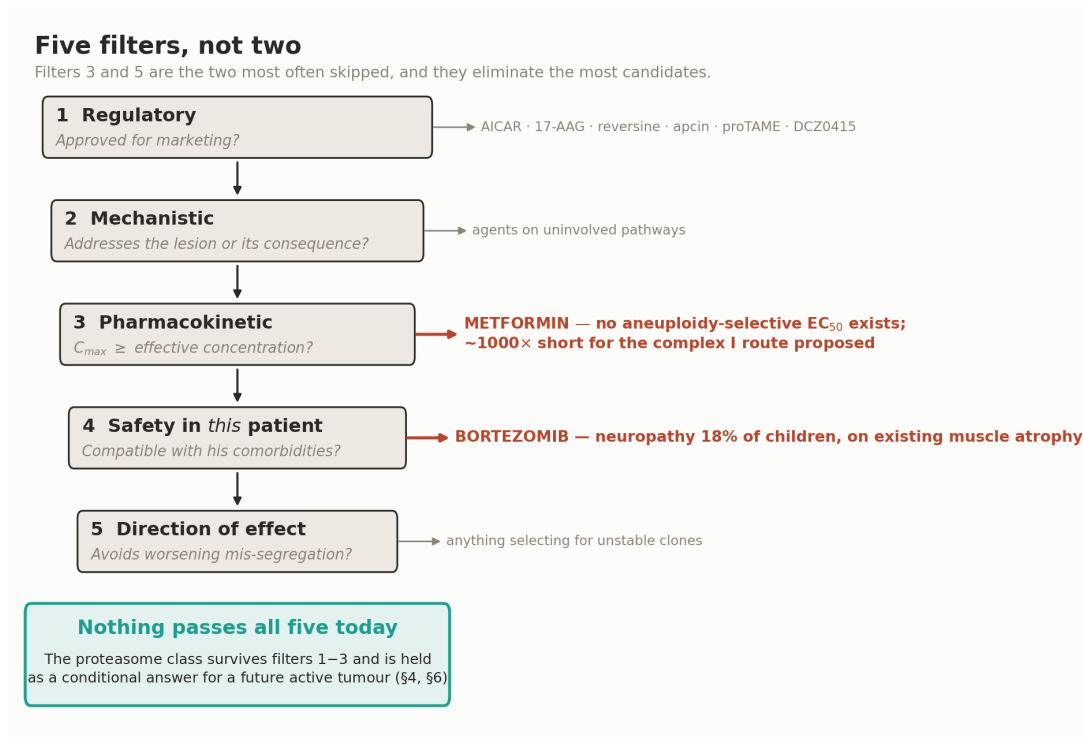
Whether the truncated transcript escapes nonsense-mediated decay is unmeasured, and should not be asserted either way without patient RNA.

At the cellular level, cell lines derived from MVA patients with biallelic mutations show “an impaired mitotic checkpoint, chromosome alignment defects, and low overall BUBR1 abundance” (PMID 20516114). BubR1 binds the B56 family of PP2A regulatory subunits through a conserved motif phosphorylated by Cdk1 and Plk1, and counteracts Aurora B at improperly attached kinetochores (Kruse et al., *J Cell Sci* 2013, PMID 23345399); the same B56 partnership opposes Aurora B to give stable kinetochore–microtubule attachment (Xu et al., *Biol Open* 2013, PMID 23789096). The region is commonly called KARD, a name neither source uses.

2. Five filters, not two

Repurposing exercises usually apply two filters: **is it approved?** and **is the mechanism plausible?** Both are necessary; neither is sufficient. Three more decide the outcome.

#	Filter	Question	What it eliminates here
1	Regulatory	Approved for marketing?	AICAR, 17-AAG, reversine, apcin, proTAME, DCZ0415
2	Mechanistic	Does it address the actual lesion or a direct consequence of it?	Agents acting on uninvolved pathways
3	Pharmacokinetic	Is $C_{max} \geq$ the effective in vitro concentration?	Metformin (~1000-fold short)
4	Patient-specific safety	Compatible with <i>this</i> child's comorbidities?	Bortezomib, at present
5	Direction of effect	Does it avoid increasing mis-segregation in survivors?	Anything selecting for unstable clones



The five-filter funnel: what each filter eliminates, and why nothing passes all five today

Filters 3 and 5 are rarely applied and eliminate the most candidates. Filter 3 is a possibility check, not a demonstration of efficacy: it ignores protein binding, exposure duration and tissue penetration, and a transient post-bolus C_{max} can overstate useful exposure.

3. Applying the filters

3.1 Metformin fails on pharmacokinetics

The chain of reasoning is genuinely attractive and we followed it to the end. Tang et al. screened isogenic trisomic mouse fibroblasts and found selective sensitivity to **AICAR**. AICAR is not approved, so it fails filter 1. Metformin is the marketed stand-in that also produces AMPK activation through complex I inhibition, and it is labelled in paediatrics from age 10. It passes filters 1 and 2.

It fails filter 3 decisively. Sakellakis (2023, PMID 37343530):

“the plasma concentration of metformin remains within the micromolar range when administered at therapeutic doses. While millimolar concentrations are necessary to inhibit complex I activity in isolated mitochondria, there is no evidence supporting the idea that metformin accumulates within the mitochondria.”

Fontaine (2018, PMID 30619086) reviews why that inhibition remains unsettled — that review exists to “gather arguments for and against each hypothesis.” In paediatrics the gap widens: dose-normalised plasma exposure varies **3.3-fold** between children, and “**renal clearance was higher in children compared to the reported data in healthy adults**”, with eGFR significantly affecting oral clearance (PMID 42399684). Metformin is renally cleared, so this child’s nephrocalcinosis makes renal function the parameter that would have to be established first — the source links clearance to eGFR, not to nephrocalcinosis specifically, and we do not stretch it further.

The distinction that decides this, stated precisely. Therapeutic metformin *does* activate AMPK in patients — that is the diabetes literature and we do not dispute it. What the micromolar-versus-millimolar gap rules out is the **complex I route at plasma concentrations**. Tang’s screening hit was AICAR, a direct AMPK agonist, so an AMPK-mediated aneuploidy-selective effect of metformin is not excluded by this argument; it is simply **untested**. No study reports a metformin aneuploidy-selective effective concentration, and the one paper reproducing AICAR’s selectivity in human cells does not test metformin. We therefore reject metformin as **unevaluable on the mechanism it is proposed for**, not as disproven

— and we say which of the two we mean, because they license different next steps: the first is answered by an experiment, the second would close the question.

We do not reject metformin out of caution. We reject it with a number — and we state what kind of number it is. For bortezomib, filter 3 is a true C_{max} / EC_{50} ratio against an EC_{50} measured in an aneuploidy-stratified model. For metformin no such EC_{50} exists: no published study gives an aneuploidy-selective effective concentration. AICAR’s aneuploidy selectivity has in fact been reproduced in human cells — a trisomy-7 colonic line is fourfold more sensitive to AICAR than its isogenic diploid counterpart (PMID 22890317). That paper does not test metformin, and no published study gives metformin an aneuploidy-selective effective concentration at all. The 1000-fold figure is therefore a comparison between *concentration ranges* — micromolar plasma against millimolar complex I inhibition — not the same quantity as the bortezomib ratio. It is a coarser instrument, and it still points the same way.

There is a deeper objection, and stating it accurately matters. Tang et al. did **not** restrict themselves to stable trisomies: Figure 6 of that paper tests AICAR and 17-AAG in *Bub1b*^{H/H} hypomorphic MEFs and in *Cdc20*^{AAA} cells — genuine ongoing-instability models with a weak checkpoint. So the screen does reach beyond a fixed extra chromosome.

What it does not reach is **metformin**, which is not tested in those models, and human biallelic BUB1B cells, which are not tested at all. The gap is therefore narrower and more specific than “stable trisomies only”: the marketed stand-in was never evaluated in the instability models, and a mouse *Bub1b*^{H/H} hypomorph is not equivalent to human p.Leu737Ter/p.Asn1002Lys.

3.2 Bortezomib passes filters 1–2, and clears filter 3 on total-drug exposure

Mechanism. Aneuploidy imposes a stoichiometric imbalance of protein complexes. Ippolito et al. (2024, PMID 39247952) dissected multiple diploid-versus-aneuploid models and found that aneuploid cells mitigate proteotoxic stress by reducing translation and

increasing protein degradation, rendering them more sensitive to proteasome inhibition. The finding was recapitulated across hundreds of human cancer cell lines and primary tumours.

An independent, orthogonal 2026 result points at the same target.

Paired genome-wide CRISPR loss-of-function screens in isogenic aneuploid versus near-euploid lines identify **proteasome subunits** among the top aneuploid-specific dependency gene groups, alongside ribosome and rRNA processing, spliceosome and mitochondrial metabolism; a focused druggable-genome arm nominates the ubiquitin-conjugating enzyme **UBE2H** as a top aneuploid-selective dependency (PMID 42094535). This is genetic loss-of-function evidence rather than pharmacology, arrived at by a different method than Ippolito's, and it converges on the ubiquitin–proteasome axis. **It is a preprint (bioRxiv) and not yet peer-reviewed**, so we weight it as corroboration, not as a second pillar. UBE2H itself has no approved inhibitor and therefore fails filter 1 today; we record it in §9 as the target to watch.

The clinically important sentence in **Ippolito et al.** is the last one: **aneuploidy level was significantly associated with the response of multiple myeloma patients to proteasome inhibitors**. That is a human signal linking degree of aneuploidy to response to this drug class, which is more than most repurposing arguments have.

We state the size of that clinical signal rather than let a reviewer find it. It rests on small cohorts: with bortezomib as a single agent, 8 patients with a complete response had significantly higher aneuploidy scores than 50 with progressive disease ($p = 0.014$, one-tailed Mann-Whitney); in a bortezomib-plus-chemotherapy-and-dexamethasone cohort, 13 complete responders versus 14 minimal responders ($p = 0.038$); a third cohort trended the same way but had only 2 non-responders. That is a genuine human association built on small numbers — enough to justify running an experiment, not enough to predict an individual patient's response.

It is also not evidence about MVA1. Myeloma is a plasma-cell neoplasm whose immunoglobulin output imposes a proteotoxic load of its own, and its aneuploidy is clonal, not variegated. MVA1 is a constitutional

aneuploidy state — a *different* setting in which the same dependency is plausible and untested. We treat the myeloma association as the reason to run the experiment, not as its result.

The numbers.

	Value	Source
EC50 in highly aneuploid lines (72 h)	< 40 nM — conservative upper bound read from Fig. 6o (5 near-euploid vs 5 highly aneuploid cancer lines, $p = 0.0317$, Mann-Whitney); the figure legend reports no numeric EC50 values, so this is read off the plot	PMID 39247952
Cmax, 1.3 mg/m ² IV, repeated dosing	89–120 ng/mL = 231–312 nM (label interval; we quote both ends)	FDA label, NDA 021602 s040, §12.3
Cmax, subcutaneous route	20.4 ng/mL = 53.1 nM	same
Filter 3 ratio (IV)	5.8 – 7.8	—

We use the upper bound of the EC50 range, and we do not treat the 2.4 nM apoptosis figure from the same paper as an EC50.

The subcutaneous route does not pass, and we withdraw the claim that it does. Its nominal ratio is 1.33 on *total* drug, and bortezomib is extensively protein-bound, so free concentration is a fraction of total Cmax. A margin of 1.33 does not survive that correction. Only the IV route clears filter 3 — and even there, the ratio is computed on total drug, which is the acknowledged weakness of the filter (§2).

3.3 Everything else: not evaluable, and we say so

Drug	Filter 3 verdict	Why
Carfilzomib	Not evaluable	No published IC50 in an aneuploidy-stratified model; Ippolito used bortezomib and MG132. Cmax is known (2.89 μ M at 56 mg/m ²) but there is no valid denominator
Ixazomib	Not evaluable	Same gap. The β 5 enzymatic IC50 (3.4 nM) is not a cellular aneuploidy-selective value
Hydroxychloroquine	Not evaluable, and mechanistically weakened	Tang's hit was <i>chloroquine</i> , not HCQ — and chloroquine did not differentially inhibit the human CIN lines in that same paper
Everolimus / sirolimus	Not evaluable, and possibly counterproductive	See §3.4

Substituting an IC50 from an unstratified tumour line would manufacture a ratio without meaning. We leave those cells empty rather than fill them.

3.4 The antagonism nobody should walk into

A team building a combination might reach for an mTOR inhibitor alongside a proteasome inhibitor: both touch protein homeostasis, so synergy feels intuitive. The evidence points the other way.

Aneuploid and CIN cells are vulnerable to proteasome inhibition **because** they carry a translational burden. Reducing protein synthesis relieves that burden — and the ovarian carcinoma work shows exactly that mechanism operating as *resistance*. Ovarian cancer cells kept their sensitivity to proteasome inhibitors in adherent culture but “**acquired resistance as spheroids ... by downregulating protein synthesis via mTORC1 suppression**”; conversely, PTEN loss drove constitutive mTORC1 activity and **resensitized** the spheroids to proteasome inhibition, in vitro and in vivo (PMID 31530568).

Two boundaries on that evidence, since we are the ones importing it. It is ovarian carcinoma spheroids, not constitutional aneuploidy. And the paper did **not** test rapalogues — the antagonism below is our inference from its mechanism, not its result.

The objection this raises against our own candidate, stated before a reviewer does. The same paper notes that this protective mechanism “may explain the overall poor response to bortezomib in clinical trials of patients with advanced ovarian cancer.” Bortezomib has therefore *failed* in at least one solid-tumour setting, and any honest reading has to carry that. Our answer is that the proposed failure mode is context-specific — a spheroid/ascites configuration suppressing translation — and that the same paper identifies a responsive subpopulation defined by mTORC1 status. It is a reason to measure the response in patient cells before believing it, which is what §6 does; it is not a reason the mechanism is wrong.

The premise of that prediction has independent 2025 support. Work in yeast shows that chromosome amplification drives premature aging through defects in ribosome quality control, that aneuploids carry an increased load of protein aggregates and signs of ubiquitin dysregulation, and — the load-bearing point for us — that the *increased translational load* is what accelerates the decline (PMID 41248159). That is a different organism and an aging phenotype rather than a drug response, so we cite it for the premise only: reducing translation relieves a burden these cells depend on carrying. It is also the reason the same axis appears twice in this report with opposite signs, since MVA is itself a progeroid syndrome (PMID 31738183).

Prediction, stated so that it can be falsified: everolimus or sirolimus would antagonise bortezomib in aneuploid cells rather than potentiate it. Any screen should test that combination explicitly and expect a negative interaction.

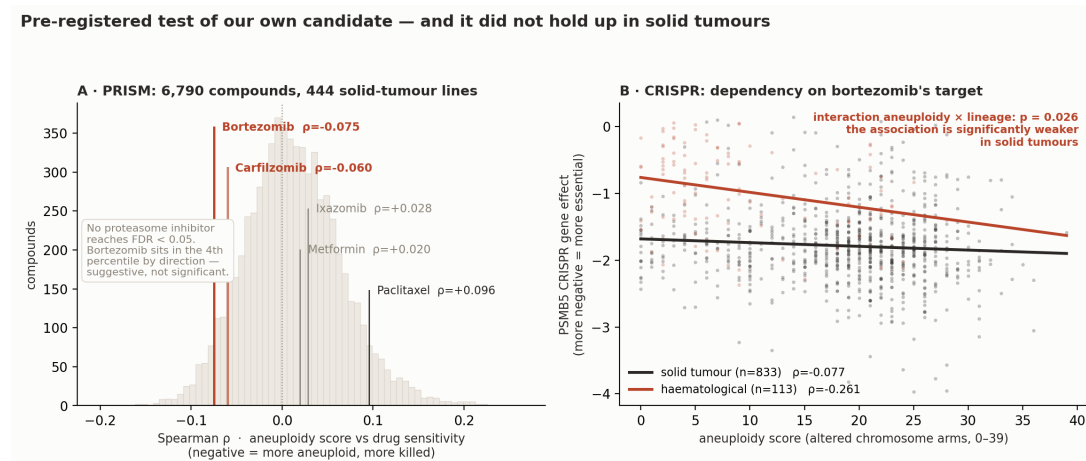
A tension we flag rather than hide. The same BubR1 mouse work reports that predisposition to sarcopenia correlates with **mTORC1 hyperactivity** (PMID 31738183) — and this child has skeletal muscle atrophy. So an independent, muscle-directed rationale for mTOR inhibition in MVA exists, and it points the opposite way from our antagonism prediction. The two are not strictly in conflict: one

concerns proteotoxic killing of aneuploid cells, the other a progeroid muscle phenotype. But anyone testing the combination should know both exist, and should not read a muscle-directed mTOR rationale as support for pairing it with a proteasome inhibitor.

3.5 We tested our own candidate on public data, and pre-registered it first

Two reviewers made the same objection to §3.2 independently: the human signal is **multiple myeloma**, and this child's tumour was an **embryonal rhabdomyosarcoma**. We had no number for solid tumours, so we produced one.

We fixed the hypotheses before we looked. The drug list, the direction of every predicted effect, the statistic, the multiplicity correction and the stopping rules were written to `depmap/PREREGISTRO.md` and committed to the public repository **before the analysis was run**; its git timestamp is the evidence. When one hypothesis turned out to be untestable mid-analysis, we recorded that as a dated amendment rather than editing the original.



Pre-registered test: our candidate among 6,790 compounds, and the lineage-restricted dependency on its target

Method. An arm-level aneuploidy score for 2,420 DepMap 24Q4 lines from absolute copy-number segments, validated against lines whose biology is independently known (chromosomally stable MSI lines average 5.0 altered arms; CIN lines 16.5). Then Spearman correlations against PRISM Repurposing drug sensitivity and DepMap CRISPR gene effect.

Result 1 — drugs. In 444 solid-tumour lines, **no proteasome inhibitor reached FDR < 0.05**. Bortezomib ($\rho = -0.075$) and carfilzomib ($\rho = -0.060$) fall in the 4th and 8th percentile of all 6,790 compounds *by direction* — suggestive and nothing more; ixazomib runs the other way. Metformin is null, as predicted. Paclitaxel, our pre-declared cytotoxic control, runs opposite, so this is not a “sick cells die more” confound. The pipeline is not blind: across all 6,790 compounds 1.1% reach $p < 0.001$ against 0.1% expected by chance.

Result 2 — the target itself. Genetic dependency on PSMB5, the subunit bortezomib binds, tracks aneuploidy strongly overall ($\rho = -0.198$, $q = 5 \times 10^{-8}$, $n = 946$) — but split by lineage it is **significant in haematological lines ($\rho = -0.261$, $q = 0.042$, $n = 113$) and not in solid lines ($\rho = -0.077$, $q = 0.118$, $n = 833$)**. The 20S core as a set is indistinguishable from a random-gene background in solid lines ($z = -0.51$), and the pre-declared negative control PSMD9 is null throughout. Adjusting for ploidy and lineage, aneuploidy still predicts PSMB5 dependency ($p = 0.0098$) while **ploidy contributes nothing** ($p = 0.88$) — this is not whole-genome doubling in disguise. The formal test of our question, the **aneuploidy $\times$ lineage interaction, is significant ($p = 0.026$)**. Solid lines are the *larger* stratum, so this is attenuation, not low power.

What it costs us, stated plainly. Two independent measurements agree: the link our bortezomib argument rests on is detectable where myeloma sits and attenuated in solid tumours. The §4 “prepared answer for a second tumour” therefore rests on evidence **we could not extend to the relevant lineage**. We do not withdraw the proposal — the direction survives for two of three proteasome inhibitors, the mechanism is unchanged, and a null across heterogeneous cancer lines is not a null in constitutional MVA. But the claim is weaker than it was before we ran this, and the experiment in §6 matters more, not less.

What we did not do. PRISM primary is a single-dose viability screen, not an EC50, and not Ippolito’s isogenic comparison. This is not a replication attempt of Ippolito and we do not present it as one. Full method, numbers and limitations: `depmap/RESULTADOS.md`.

4. Filter 4: safety in *this* child — where the candidate stops

Bortezomib clears the pharmacology. It does not clear this patient today.

Concern	Data	Relevance here
Peripheral neuropathy	25/140 (18%) of paediatric patients, motor neuropathy 11/140 (8%) (FDA BPCA Clinical Review, NDA 021602, 2015). In adults, grade ≥ 2 in 39% and grade ≥ 3 in 15% by IV route (FDA label, NDA 206927 s002, 2022 — a different source from the PK figures above)	This child has skeletal muscle atrophy (HP:0003202). Superimposing a motor neuropathy on reduced neuromuscular reserve is the decisive objection
Thrombocytopenia, neutropenia, infection	FDA label	Prior chemotherapy for ERMS; marrow reserve unknown
Gastrointestinal toxicity, hypotension	FDA label	Failure to thrive (HP:0001508) — anything suppressing intake is costly
Renal	PK not clinically altered by renal impairment per label	Nephrocalcinosis (HP:0000121) still requires renal function, hydration and electrolytes to be established first

Bortezomib is therefore not proposed as prophylaxis or as chronic therapy for this child.

But filter 4 is patient-specific, not universal — and that is the point. Read what follows against §3.5: our own pre-registered test could not extend the proteasome dependency to solid-tumour lineages, and an embryonal rhabdomyosarcoma is a solid tumour. The argument below is weaker than the mechanism alone suggests, and the ex-vivo test in §6 is the gate, not a formality. For a patient with MVA **and an active malignancy**, where cytotoxic therapy is on the table regardless and the comparator is conventional chemotherapy rather than nothing, the balance inverts. In that setting a proteasome inhibitor is not an exotic addition: it is an approved agent that targets the best-characterised downstream consequence of the underlying lesion — proteotoxic

dependency — supported by a human clinical signal tying response to aneuploidy burden. It does not correct the checkpoint defect, and we do not claim it does.

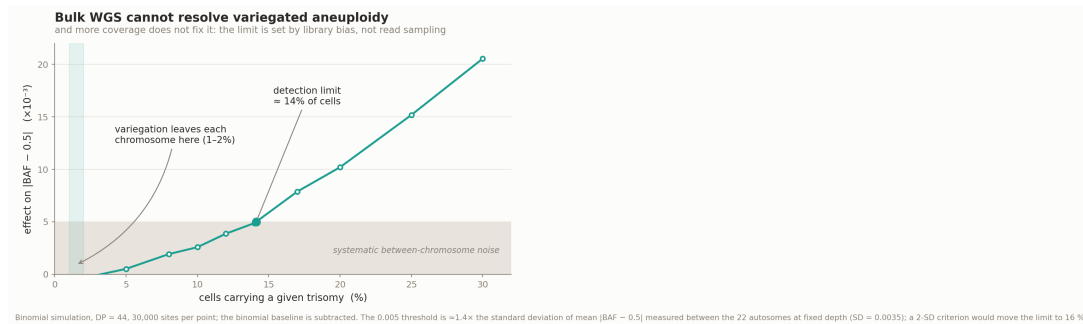
This child survived an embryonal rhabdomyosarcoma, and the risk is gene-specific: “individuals with biallelic TRIP13 or BUB1B mutations have a high risk of embryonal tumors” (PMID 28553959), and an E-RMS arising in an MVA case with biallelic BUB1B has been characterised directly (PMID 16182441). **The realistic clinical question is not “what do we give him today” but “what do we reach for if there is a second tumour” — and that question deserves an answer prepared in advance, with its evidence and its threshold, rather than improvised under pressure.**

There is a second, immediate implication. Spindle poisons — vincristine, taxanes — depend on a competent mitotic checkpoint. With reduced BubR1 that arrest is less robust; a case of rhabdomyosarcoma with PCS/MVA treated with **reduced-intensity chemotherapy** is on record (PMID 31184400 — a single case, not a guideline and not a safety estimate; PubMed carries no abstract for it, so we cite only what its title states and do not name the regimen). This is a **relative** contraindication weighed against probability of cure, never a reason to withhold curative therapy. If a mitotic-poison-sparing regimen is ever needed, there is **no validated replacement agent**: a proteasome inhibitor would be a bench hypothesis to test first, not a substitution anyone could make at the bedside.

We state the boundary explicitly: this is a hypothesis for a tumour board, not a prescription.

5. Why any endpoint must be measured per cell

Any trial of the above needs a readout for **aneuploidy burden**. We asked whether the data this hackathon provides could supply one, and computed the answer instead of asserting it.



Detection limit of bulk WGS for mosaic aneuploidy

We are not proposing a new assay. The correct endpoints exist and are standard: metaphase karyotyping with premature chromatid separation scoring is the diagnostic criterion for MVA; the micronucleus assay is an international standard (OECD TG 487); and single-cell DNA sequencing has been used to assay chromosome copy number genome-wide since Knouse et al. 2014 (PMID 25197050). We cite that work for the method, not for our conclusion — its own finding is that aneuploidy is *rarer* in liver and brain than previously reported.

What we contribute is the number.

The analysis used the **VCF only** — no BAM, hence no GC-LOESS normalisation. Four analyses were run, of which three are refinements of the same B-allele-frequency statistic and one (depth-based dosage) is orthogonal; they are not four independent tests. In a trisomy, heterozygous sites split symmetrically, so the *mean* BAF does not move and the correct statistic is $\text{mean } |BAF - 0.5|$.

The measured standard deviation of $\text{mean } |BAF - 0.5|$ across the 22 autosomes at fixed depth (DP 40–48, 857,435 sites) is **0.0035**, so the 0.005 noise floor we use — arising from GC content, mappability and segmental duplications — is a ≈ 1.4 -SD criterion. Simulating at DP 44 against it gives a **practical detection limit of a clonal whole-chromosome trisomy present in ~14% of cells**; a conventional 2-SD criterion moves it to 16% and 3-SD to 20%. Every stricter choice makes the assay look worse, so the conclusion does not rest on the threshold.

Read sampling is not what sets that limit. The simulation subtracts the binomial baseline at DP 44, so the 0.005 threshold a signal must clear is the *measured dispersion between chromosomes* — systematic library bias, not counting noise. The depth-based dosage test points the same way by a separate route: Poisson error when averaging a whole

chromosome at 44× is ~0.03%, far below the observed deviations. These are two different statistics and we do not conflate them; both imply that sequencing deeper would barely move the limit.

As an **illustrative bound, not a measurement of this child**: if 30% of cells were aneuploid — the order of magnitude of the diagnostic criterion, not a figure from this patient — spread across 22 autosomes, each chromosome would be affected in **1–2% of cells**, and gains and losses of the same chromosome cancel in bulk. That sits **7–16× below** the 14–16% limit. And **bulk averaging erases variegation by construction**, at any depth. The cytogenetic hallmark of MVA, premature chromatid separation, leaves no trace in DNA sequence at all.

Every candidate signal we did observe traced to coverage, to segmental duplications (22q11, centromeres), or partially to GC content — partially, because our measured GC–coverage correlation was $r = 0.43$ from a sparse sample, and chr21, the largest deviation, is GC-poor. That is an insufficient measurement rather than an established explanation, and we report it as such.

Consequence: no proposal whose endpoint is “reduced aneuploidy burden” can be evaluated with these data. Any screen must measure per-cell mis-segregation.

6. The go/no-go experiment — an eight-week assay window

Scoped to be executable by one laboratory in one window, not as a grant application.

Question. Are BUB1B-deficient patient cells preferentially sensitive to bortezomib at clinically achievable concentrations, and does that sensitivity come without increasing mis-segregation among the survivors?

What this experiment can and cannot decide, stated first. Dermal fibroblasts from this child are *constitutional* MVA cells, not tumour cells. A sensitivity difference between them and unaffected controls therefore measures whether the proteotoxic dependency transfers from cancer aneuploidy to constitutional aneuploidy at all — it does

not measure tumour selectivity, and on its own it is as compatible with systemic toxicity as with therapeutic promise. That is why the go criterion below is *not* “patient cells are more sensitive”. It is also why \$5 matters here: variegation leaves each chromosome affected in 1–2% of cells, so a fibroblast population may simply not be aneuploid enough to express a dependency that Ippolito measured in *highly* aneuploid lines. **A negative result is therefore ambiguous by design** — it fails to support transfer, but it does not close the tumour-board question, which would need aneuploidy-high tumour-derived material. We say this now rather than after the data.

Week	Step	Readout
1–3	Patient dermal fibroblasts from skin biopsy, plus two matched controls. Patient cells are required: RPE1 with reversine-induced aneuploidy reproduces Ippolito’s own system and cannot test whether the dependency transfers to BUB1B-deficient constitutional MVA — we list it as a positive control, not a substitute. Without patient cells the go/no-go question is not answerable	Growth; baseline karyotype
2–4	Allele fate: allele-specific RT-PCR ± NMD inhibitor; quantitative western blot with N- and C-terminal antibodies	N+/C– band implies truncation; loss of both implies NMD
4–6	Bortezomib dose–response, 0–400 nM with dense sampling below 100 nM, 72 h, patient versus control versus an aneuploidy-high positive control (reversine-treated RPE1)	EC50 and the ratio between them. The range spans the 312 nM clinical Cmax so the advancement threshold falls inside the data
4–6	Combination arm: bortezomib + everolimus	Tests the predicted antagonism
6–8	Micronucleus assay (OECD TG 487) on survivors of every condition, plus FISH for 3–5 chromosomes	Mis-segregation rate — filter 5

Timeline, stated honestly in the heading it belongs in. Eight weeks is the *assay* programme, and it holds only once a patient fibroblast line exists. Establishing $15\text{--}20 \times 10^6$ fibroblasts from a fresh biopsy typically takes 4–8 weeks on its own, so from a standing start the realistic figure is **about twelve weeks**, not eight. We use “eight-week experiment” as shorthand for the assay window and nowhere as a promise of a twelve-week answer.

Advancement criteria, fixed before the data. We state them in the direction that avoids rewarding toxicity:

1. **A dependency signal, not merely a sensitivity difference:** an EC50 shift between patient and control fibroblasts whose 95% confidence interval excludes 1, of a magnitude at least equal to the near-euploid-versus-highly-aneuploid contrast in Ippolito Fig. 6o. A shift smaller than that is not the effect we predicted.
2. **Effect at $\leq 312\text{ nM}$,** the upper end of the clinical Cmax interval.
3. **No increase in micronuclei among survivors** — filter 5.
4. **A stop signal, not a go signal:** if patient fibroblasts are markedly more sensitive than controls *without* the aneuploidy-high positive control showing a larger shift still, the most likely reading is constitutional toxicity in an MVA patient rather than selective killing of aneuploid cells. That outcome stops the programme too, and it is the outcome we would most easily have mistaken for success.

Failing any of 1–3, or triggering 4, stops the programme.

Falsifying results, stated at the precision the design supports. If patient fibroblasts are no more sensitive than controls, the experiment **fails to support transfer of the dependency into these cells** — it does not falsify a dependency in aneuploidy-high tumour material, which is the setting §4 actually proposes and which this design does not sample. That distinction is the difference between “we did not see it here” and “it is not there”, and we hold ourselves to the first. If bortezomib raises micronucleus frequency among survivors, that **does** falsify the proposal outright: it fails filter 5 regardless of how well it kills.

The stage this design does not contain. Deciding tumour selectivity requires aneuploidy-high tumour-derived material with a near-euploid comparator, which no eight-week window starting from a skin biopsy can supply. We treat the fibroblast arm as the gate that decides whether that second stage is worth building, not as a substitute for it.

This design can fail, which is the point of proposing it.

7. Actionable now, without any drug

The intervention that most changes this child's prognosis today is not a molecule.

There is a published consensus for MVA specifically, and it should be the anchor. The 2024 update from the AACR Childhood Cancer Predisposition Workshop covers mosaic variegated aneuploidy by name and states: *“This group recommended renal ultrasound surveillance every 3 months from birth until age 7, for all MVA conditions including those with an unknown genetic cause”,* together with *“regular clinical assessment including review of systems to identify signs of rhabdomyosarcoma and other malignancies”,* avoidance of radiation exposure, and HPV vaccination (PMID 39264246). The same source records that cancers in MVA1 are *“predominantly Wilms tumor and rhabdomyosarcoma, but MDS, AML and ALL have also been reported”*.

We had reached the 3-monthly-to-age-7 schedule independently, by extension from UK Wilms-risk recommendations that stop at age 5 (PMID 16857697). We report that convergence rather than our derivation: **the schedule below is consensus guidance, not our inference**, and a family should be given it as such.

Where the consensus leaves a gap, and it is the gap that matters for this child. The window closes at seven. A 2026 case report describes an MVA3 girl whose Wilms tumour appeared at three and whose orbital embryonal rhabdomyosarcoma appeared at **twelve** (PMID 42595739) — five years past the end of renal-ultrasound surveillance, at a site renal ultrasound does not image. Renal ultrasound is a kidney-directed test; rhabdomyosarcoma is not a kidney tumour, which is why the consensus pairs it with clinical

assessment rather than with imaging. For a child who has *already had* an embryonal rhabdomyosarcoma, we therefore propose that the clinical-examination arm continue past seven, and we mark it as our extension rather than as guidance.

Surveillance windows must be anchored to the child's current age, which is not in the dataset and must be confirmed before use. If the age is unknown, that is the first thing to establish — nothing below is actionable without it. This is a discussion framework for an expert cancer-predisposition clinic, not a substitute for one:

Period	Surveillance	Frequency
Birth to age 7	Renal ultrasound (we suggest extending the field to liver, retroperitoneum and pelvis)	Every 3 months — consensus recommendation for all MVA conditions (PMID 39264246). The wider abdominal field is our addition
To age 10, then reviewed	Full paediatric and oncological examination: abdomen, nodes, head and neck, orbit , skin, genitalia, soft tissues	3-monthly to age 7, then 6-monthly. The consensus asks for clinical assessment for rhabdomyosarcoma without setting an end date; continuing past 7 is our extension , prompted by the age-12 orbital ERMS in PMID 42595739. Catches superficial ERMS that renal ultrasound cannot
Post-ERMS	Relapse follow-up per tumour protocol	Coordinate to avoid duplicate imaging
Through childhood	Full blood count with differential	6-monthly is reasonable given reported leukaemia, but no evidence of benefit exists
Lifelong	Family education; annual predisposition clinic	Urgent review for mass, haematuria, abdominal distension, persistent pain, neurological deficit, bleeding or weight loss

If the child is already past 7, missed scans should not be “caught up”: a baseline assessment and an individualised decision by the predisposition team is the right course.

We do **not** propose serial AFP, repeat CT, PET-CT or routine whole-body MRI while asymptomatic — no demonstrated benefit in MVA, weighed against sedation, false positives, cost and radiation. The consensus is explicit that **radiation exposure should be avoided** in these disorders (PMID 39264246), which is an argument against imaging-heavy surveillance and, separately, a consideration for any future treatment plan.

One implication for the parents, which the phenotype document already hints at. The clinical document lists recurrent spontaneous abortion (HP:0200067) — an obstetric history, not a feature of the child. If the two variants are in *trans* — which is the interpretation this report adopts but does not demonstrate (§10) — each parent would be expected to carry one of them; segregation testing is what would establish that, and it should precede any carrier-specific interpretation. A 2026 study of two unrelated families with unexplained recurrent pregnancy loss identified novel heterozygous *BUB1B* variants and found **significantly elevated premature chromatid separation rates in the carriers' lymphocytes** by G-banding and centromere FISH, with a trend toward reduced BUBR1 (PMID 42434306). Those variants were classified VUS and the cohort is two families, so this is a hypothesis, not a finding about this family. But it is cheap to act on and it is squarely in the family's interest: a PCS assay on parental lymphocytes and formal genetic counselling would test whether the miscarriage history is itself part of the carrier phenotype, and would inform any future pregnancy. It is cheap, it can be done this month **alongside** the consensus surveillance above, and it replaces no part of it.

8. What we do not propose

- **MPS1/TTK inhibitors and reversine** — not approved, and they weaken a checkpoint that is already insufficient. Reversine is a research tool for *inducing* aneuploidy, the opposite of the goal.
- **Apcin, proTAME** — APC/C tool compounds, not marketed.
- **DCZ0415** — experimental TRIP13 inhibitor: wrong gene, not approved.
- **AICAR and 17-AAG** — genuine hits in the Tang screen, neither marketed. Useful as in vitro positive controls only.

- **Metformin** — rejected at filter 3 as unevaluable on the mechanism proposed for it (§3.1).
 - **Senolytics** — the BubR1 senescence evidence rests on *genetic* clearance of p16^{Ink4a} cells via the INK-ATTAC transgene (PMID 22048312), which is not a transferable drug; no agent is approved with a senolytic indication; and senescence is an anti-tumour barrier worth preserving in a cancer-predisposition syndrome.
 - **mTOR inhibitors combined with proteasome inhibition** — predicted antagonism (PMID 31530568).
 - **Bortezomib for this child today** — fails filter 4, as set out in §4.
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9. Scalability

- The **five-filter framework**, particularly filters 3 and 5, applies to any repurposing exercise in any rare disease. It is the transferable product of this work.
- **The MVA genes are not interchangeable.** MVA2 arises from biallelic *CEP57* (PMID 21552266) and behaves differently from MVA1 and MVA3 on exactly the axis that matters here — see the CEP57 caveat below.
- The **proteotoxic vulnerability**, if §6 shows it transfers, would extend most plausibly to **TRIP13**-related MVA, which shares both the tumour risk and the severity of checkpoint failure: “individuals with biallelic TRIP13 or BUB1B mutations have a high risk of embryonal tumors ... their cells display severe SAC impairment” (PMID 28553959) — and the 2026 case in §7 is a TRIP13 patient with two embryonal tumours (PMID 42595739). **It should not be assumed for CEP57.** Yost reports that CEP57-related MVA “is not associated with embryonal tumors” with cells showing “minimal SAC deficiency”, and the 2024 consensus is blunter still: **“At the time of writing, none of the 15 individuals with MVA2 described in literature developed cancer”** (PMID 39264246). Less mis-segregation should mean less proteotoxic burden, so the rationale is weaker there. No proteasome data exist for either, and §10 states that transfer to MVA1 itself is unproven.
- **The target to watch is UBE2H.** The 2026 paired CRISPR screens nominate it as a top aneuploid-selective dependency linked to mitochondrial proteostasis (PMID 42094535, preprint). It has no

approved inhibitor, so it fails filter 1 today and is not a candidate — but it is where an aneuploidy-selective agent would most plausibly come from next, and a repurposing exercise repeated in two years should start there rather than at metformin.

- **A reusable worksheet** is included as the Appendix: the five filters as blank rows, with the metformin and bortezomib verdicts as the worked example, so a team working on a different rare disease can apply the framework without reading this report.
 - The **gene panel** (BUB1B, CEP57, TRIP13, CENATAC, MAD1L1, MAD2L1BP, CEP192, BUB1, SMC5, TRIM37, CENPE) and the artefact controls used in the mosaicism analysis transfer to any MVA or PCS workup.
 - The **micronucleus → scDNA-seq endpoint** applies to mitotic CIN disorders. It does **not** transfer to Fanconi anaemia or Bloom syndrome, whose diagnostic assays are DEB/MMC chromosome breakage and sister-chromatid exchange respectively — a different lesion, not substitutable.
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10. Limitations

- **Phase is not established, and we measured that it is not establishable from these data.** There are no parental samples, and the challenge distributes a VCF with no alignments, so no read-backed phaser can be run at all; the two variants sit **10,911 bp** apart and the caller emitted no phase tag for either. Reference-panel phasing fails for a separate and quantifiable reason: inside that interval the child carries exactly three heterozygous sites — the two causal variants and `chr15:40216470` — and **all three are absent from the 1000 Genomes 30× phased panel** (3,202 individuals); the nearest panel-present heterozygous markers are 16.8 kb upstream and 809 bp downstream, with nothing between them to link the alleles. A Beagle 5.5 run against that panel keeps 1,391 of 1,452 window variants and drops all three that matter. The absence is what the frequencies predict: expected copy counts in 6,404 haplotypes are 0.64 for the nonsense and **0.006** for the missense. Two heterozygous variants in a singleton are a priori 50/50 cis or trans, and rarity does not phase them. What supports *trans* is that the child is affected, that MVA1 is recessive, that

truncating-plus-missense is the recurrent pattern, and that the phenotype matches. **It remains a diagnosis pending segregation, and parental genotyping at the two loci is the cheapest experiment in this report.**

- **The retrieval pipeline is itself insensitive to phase.** In a controlled test on public data, Exomiser 15.1.0 reads phase into its ACMG evidence (trans → PM3, cis → BP2) but not into the compound-heterozygous model: two pathogenic nonsense alleles declared on the *same* chromosome are still called AR_COMP_HET, still classified PATHOGENIC and still ranked first, at an absolute combined-score difference of 0.0034. Nothing in the Track 1 retrieval would have flagged a *cis* configuration.
- **p.Asn1002Lys has no functional assay**, and its apparent ClinVar entry is for a different nucleotide.
- **The bortezomib EC50 is a conservative upper bound** read from a figure panel rather than a tabulated value, and the Cmax is a label interval rather than a point estimate; the ratio of 5.8–7.8 should be read as an order-of-magnitude statement computed on total, not free, drug.
- **Ippolito et al. studied cancer aneuploidy**, not constitutional variegated mosaicism. Whether the proteotoxic dependency transfers is precisely what the proposed experiment tests; we have not assumed it.
- **No trial specific to MVA1** was found in ClinicalTrials.gov, Open Targets, ChEMBL or CMap/LINCS. Absence of an interface result is not proof of absence.
- **The child's current age is unknown to us**, and the surveillance schedule depends on it.

A closing note

The family of this child published their son's genome so that strangers might help. They will read what comes back.

We have tried to write something useful rather than something reassuring. The honest answer to “what drug should he take today” is none — and we show why the obvious candidate cannot work rather than proposing it softly. The useful answer is a mechanism that

matches his disease, an approved drug that reaches the required concentration, an explicit statement of the condition under which it would become appropriate, and an experiment — an eight-week assay window, about twelve weeks from a fresh biopsy — that could fail to support it in this child's own cells. That experiment would not close the tumour-board question on its own, and §6 says so.

That is not a cure. It is a prepared answer to a question this family may unfortunately have to ask.

Appendix — The five-filter worksheet (reusable)

The transferable product of this work. Blank rows for a new disease; the two decided rows as worked examples. A candidate must pass all five. Filters 3 and 5 eliminate the most candidates and are the two most often skipped.

Candidate	1. Approved?	2. Addresses lesion or a direct consequence?	3. Cmax ≥ effective concentration?	4. Safe in <i>this</i> patient?	5. Does not increase mis-segregation among survivors?	Verdict
Metformin	Yes (paediatric label from age 10)	Proposed as the marketed stand-in for the AICAR/ AMPK hit	Not evaluable for aneuploidy selectivity — no such EC50 exists; and ~1000× short for the complex I mechanism proposed (µM plasma vs mM required)	Renal clearance; nephrocalcinosis unassessed	Not reached	Rejected at filter 3 as unevaluable on the proposed mechanism
Bortezomib	Yes	Yes — proteasome dependency of aneuploid cells	Yes on total drug; 231–312 nM Cmax vs EC50 <40 nM (5.8–7.8×); IV only, not subcutaneous	No — neuropathy in 18% of children, on pre-existing muscle atrophy	No evidence it increases mis-segregation; to be tested	Fails filter 4 for this child; conditional for a patient with an active tumour
(your candidate)						
(your candidate)						

How to fill it in. Filter 3 needs a real denominator: an effective concentration measured in a model stratified for the disease mechanism. If none exists, write *not evaluable* — do not substitute an IC50 from an unrelated model, because that manufactures a ratio with no meaning. Filter 4 is patient-specific by construction, so a “fail” here is a statement about one person and one moment, not about the drug. Filter 5 asks the question repurposing exercises forget: among the cells that survive the drug, is the underlying defect made worse?

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